Orthogonal complements

Orthogonal complement: for a subspace $U \subseteq \mathbb{R}^n$, $U^\perp = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \cdot \mathbf{y} = 0 \text{ for all } \mathbf{y} \in U\}$.

Properties: $U^\perp$ is a subspace; $\{\mathbf{0}\}^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \{\mathbf{0}\}$. If $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, then $U^\perp = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \cdot \mathbf{x}_i = 0 \text{ for } i = 1, \dots, k\}$.

Projection

Orthogonal projection: for $\mathbf{x} \in \mathbb{R}^n$ and a subspace U, $\mathbf{p} = \text{proj}_U(\mathbf{x}) \in U$ is the unique closest vector: $\|\mathbf{x} - \mathbf{p}\| < \|\mathbf{x} - \mathbf{z}\|$ for every $\mathbf{z} \in U$ with $\mathbf{z} \neq \mathbf{p}$.

Projection theorem: if $\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$ is an orthonormal basis of U, then

$$\mathbf{p} = \text{proj}_U(\mathbf{x}) = \sum_{i=1}^{m} (\mathbf{x} \cdot \mathbf{q}_i)\mathbf{q}_i, \qquad \mathbf{v} = \mathbf{x} - \mathbf{p} \in U^\perp.$$

The vector $\mathbf{v}$ is the **residual**; if $\mathbf{x} \in U$, then $\mathbf{p} = \mathbf{x}$ and $\mathbf{v} = \mathbf{0}$. If $Q = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_m]$, then $\mathbf{p} = QQ^T\mathbf{x}$.

Normalization: the formula requires an orthonormal basis. For a nonzero orthogonal spanning set $\{\mathbf{f}_i\}$, use $\mathbf{q}_i = \mathbf{f}_i / \|\mathbf{f}_i\|$.

Subtracting components: for any orthonormal set $\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$ and $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{v} = \mathbf{x} - \sum_i (\mathbf{x} \cdot \mathbf{q}_i)\mathbf{q}_i$ is orthogonal to each $\mathbf{q}_i$.

Gram-Schmidt orthonormalization

Expanding an orthonormal set: if $\mathbf{x}$ lies outside the span of an orthonormal set $\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$, the residual $\mathbf{v}$ above is nonzero. Set $\mathbf{q}_{m+1} = \mathbf{v}/\|\mathbf{v}\|$. The enlarged set is orthonormal and

$$\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_{m+1}\} = \text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_m, \mathbf{x}\}.$$

Gram-Schmidt algorithm: for $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ with linearly independent columns, for $j = 1, \dots, n$ set

$$r_{ij} = \mathbf{q}_i^T \mathbf{a}_j \ (i < j), \quad \mathbf{p}_j = \sum_{i<j} r_{ij}\mathbf{q}_i, \quad \mathbf{v}_j = \mathbf{a}_j - \mathbf{p}_j, \quad r_{jj} = \|\mathbf{v}_j\|, \quad \mathbf{q}_j = \mathbf{v}_j / r_{jj}.$$

Here $\mathbf{p}_1 = \mathbf{0}$. If $r_{jj} = 0$, $\mathbf{a}_j$ lies in the span of the earlier columns and no new unit direction is produced.

Output theorem: for every $k = 1, \dots, n$, the $\mathbf{q}_i$ through step k form an orthonormal basis of $U_k = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$, and

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\} = U_k.$$